AI Automation & Agents

Build Workflows. Deploy Agents. Work Smarter.

CENTER FOR ADVANCED RESEARCH COMPUTING, UNIVERSITY OF NEW MEXICO

Course syllabus

Non-credit professional development from the Center for Advanced Research Computing at the University of New Mexico. Five modules, 40 hours, self-paced.

Instructor of record
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Course site: https://tyson-swetnam.github.io/AI-Automation-and-Agents/

Course information

Parameter	Details
Format	5 modules, self-paced online, rolling enrollment, with optional instructor-led cohort sessions
Total hours	40 hours (8 hours per module)
Level	Intermediate
Audience	Workforce professionals and graduate students seeking applied AI skills
Time commitment	About 8 hours per module
Tools used	Claude Desktop (Cowork + Code), n8n, LangChain, LangGraph, CrewAI, LangSmith, Ollama, Openwork, Google Colab
Assessment	Concept quizzes, guided lab exercises, one hands-on project per module, case study analyses, peer reviews, and a capstone deployment policy
Prerequisites	Basic Python knowledge
Credit	Non-credit professional development; a digital certificate of completion is awarded
Free-tier path	A no-cost alternative track is available using Ollama, n8n Community Edition, and Google Colab

Course description

This course moves beyond individual AI prompts into the world of AI agents and automated workflows: systems that can take actions, use tools, manage files, and complete multi-step tasks on your behalf. Learners progress from foundational agent theory through to structured agent frameworks used in industry, gaining transferable skills applicable to any organization or platform.

The course follows a layered progression. Module 1 builds the conceptual foundation: defining what agents are, surveying the no-code, low-code and code-first automation spectrum, and introducing hands-on agent interaction through a local model (Ollama) and agent interface (Openwork). Subsequent modules introduce the leading open-source agent frameworks (LangChain, LangGraph and CrewAI) through guided Jupyter Notebook labs on Google Colab, covering reasoning architectures, tool integration, memory, retrieval-augmented generation, and multi-agent coordination. Throughout, the emphasis remains practical and workforce-relevant.

The course closes with a dedicated module on responsible agentic AI, covering production deployment, evaluation, observability, security (the OWASP Top 10 for LLM Applications), and governance grounded in the EU AI Act and the NIST AI Risk Management Framework (AI RMF 1.0).

Course learning outcomes

Upon successful completion of this course, learners will be able to:

1. **Design** an AI agent system specification that selects and justifies the reasoning architecture, memory configuration, retrieval pipeline, and multi-agent coordination pattern for a given task and deployment context.
2. **Implement** end-to-end AI agent pipelines integrating tool use, vector retrieval, conversational memory, multi-agent workflows, and observability instrumentation using current industry frameworks.
3. **Diagnose** failures in AI agent systems by analyzing reasoning traces, RAGAS evaluations, coordination logs, and observability data, attributing deficiencies to specific architectural causes.
4. **Evaluate** an AI agent system's production readiness across accuracy, cost, latency, safety, and coordination efficiency, and determine when multi-agent complexity is warranted.
5. **Apply** responsible AI governance frameworks (EU AI Act, NIST AI RMF) to classify risk, specify mitigations and human oversight checkpoints, and produce accountability documentation.

Course overview

#	Module	Key topics	Hours
1	**From Prompts to Pipelines: Thinking Like an Automation Designer**	Agent loop; no-code/low-code/code-first spectrum; workflow decomposition; Ollama and Openwork; key vocabulary	8
2	**Agent Reasoning Architectures and Tool Integration**	Chain-of-Thought, ReAct, Tree of Thoughts, LATS; tool integration with LangChain; prompt engineering; trace evaluation; architectural trade-offs	8
3	**Total Recall: Memory Architectures and Retrieval-Augmented Generation**	Parametric vs. non-parametric memory; six-stage RAG pipeline; conversational memory types; retrieval optimization; RAGAS evaluation	8
4	**Multi-Agent Systems**	Coordination architectures; LangGraph multi-agent implementation; three-role pipeline design; CrewAI vs. LangGraph comparison; coordination failure diagnosis	8
5	**Responsible Agentic AI: Production Deployment, Evaluation, and Responsible AI**	Technical debt and production infrastructure; multi-dimensional evaluation and CI/CD; observability with LangSmith; OWASP Top 10; EU AI Act, NIST AI RMF, human-in-the-loop	8

Completion and certification

This course is **non-credit professional development**. It does not carry university credit, and completing it does not enrol you in a degree program.

- Complete all five module projects and pass all five concept quizzes (70% or above) to earn a digital certificate of completion.
- A free-tier completion path is supported: learners who complete all activities using Ollama, n8n, and Google Colab in lieu of paid Claude tools are eligible for the same certificate.
- Bring your own work: apply course projects to real tasks from your job or studies. The Workflow Audit and the capstone deployment plan are portfolio-quality artifacts.

Grade weights for each module are stated in that module's overview; see How this course works for the pattern.

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